ANDROID PROGRAMMING PRACTICAL 15

Module 1 · RecyclerView modern list

Aim

Implement RecyclerView as a modern alternative to ListView.

Project name: RecyclerViewDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create an Empty Views Activity project and add the AndroidX RecyclerView dependency if not already available.

2. Add RecyclerView to activity_main.xml.

3. Create a row layout such as item_student.xml.

4. Create a data model class for each row.

5. Implement a RecyclerView.Adapter and ViewHolder; bind model data in onBindViewHolder().

6. Set a LinearLayoutManager and adapter on the RecyclerView.

7. Run the app, scroll the list, and verify view recycling and any item-click handling.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

A scrollable RecyclerView displays rows through a custom adapter and ViewHolder.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.